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1 Law of Large Number

1.1 Independence

Definition 1.1.1 ▶ Independence

- Two events A and B are ***independent*** if $P(A \cap B) = P(A)P(B)$.
- Two random variables X and Y are ***independent*** if for all $C, D \in \mathcal{R}$,

$$P(X \in C, Y \in D) = P(X \in C)P(Y \in D)$$

  that is, the events $A = \{X \in C\}$ and $B = \{Y \in D\}$ are independent.
- Two σ-fields $\mathcal{F}$ and $\mathcal{G}$ are ***independent*** if for all $A \in \mathcal{F}$ and $B \in \mathcal{G}$ the events A and B are independent.

Exercise 1.1.2

1. Show that if A and B are independent, then so are A^c and B, A and B^c, and A^c and B^c.
2. Conclude that events A and B are independent if and only if their indicator random variables 1_A and 1_B are independent.

Proof. 1. If A and B are independent, then

$$P(A \cap B) = P(A)P(B)$$

Note that

$$P(A \cap B) + P(A^c \cap B) = P(B)$$

We have

$$P(A^c \cap B) = P(B)(1 - P(A)) = P(B)P(A^c)$$

Similarly,

$$P(A \cap B^c) = P(A)P(B^c)$$

$$P(A^c \cap B^c) = P(B^c) - P(A \cap B^c) = P(B^c) - P(A)P(B^c) = P(A^c)P(B^c)$$

2.

$$\sigma(1_A) = \{\varnothing, A, A^c, \Omega\}, \quad \sigma(1_B) = \{\varnothing, B, B^c, \Omega\}$$

If $\sigma(1_A)$ and $\sigma(1_B)$ are independent, it is clear that A and B are as well. Conversly, only need to show that $\varnothing$ and A are independent, the remaining follows from 1. and symmetry. We have

$$P(\varnothing \cap A) = 0 = P(\varnothing) P(A)$$

□

Definition 1.1.3

- σ-fields $\mathcal{F}_1, \mathcal{F}_2, \dots, \mathcal{F}_n$ are ***independent*** if whenever $A_i \in \mathcal{F}_i$ for $i = 1, \dots, n$, we have
$$P\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} P(A_i)$$
- Random variables $X_1, \dots, X_n$ are ***independent*** if whenever $B_i \in \mathcal{R}$ for $i = 1, \dots, n$ we have
$$P\left(\bigcap_{i=1}^{n} \{X_i \in B_i\}\right) = \prod_{i=1}^{n} P(X_i \in B_i)$$
- Sets $A_1, \dots, A_n$ are ***independent*** if whenever $I \subset \{1, \dots, n\}$ we have
$$P\left(\bigcap_{i \in I} A_i\right) = \prod_{i \in I} P(A_i)$$

Remark.
The last definition looks slightly different from the former two, but it is consistent with the independence of the indicator variables $1_{A_i}, 1 \leq i \leq n$.

Exercise 1.1.4

Let $A_1, A_2, \dots, A_n$ be independent. Show
1. $A_1^c, A_2, \dots, A_n$ are independent;
2. $1_{A_1}, \dots, 1_{A_n}$ are independent.

Proof. 1. If $1 \notin I$, then it is obvious that

$$P\left(\cap_{i\in I} A_i\right) = \prod_{i\in I} P\left(A_i\right).$$

If $1 \in I$, note that

$$\begin{aligned} P\left(\cap_{i\in I\setminus\{1\}} A_i \cap A_1^c\right) &= P\left(\cap_{i\in I\setminus\{1\}} A_i\right) - P\left(\cap_{i\in I\setminus\{1\}} A_i \cap A_1\right) \\ &= \prod_{i\in I\setminus\{1\}} P\left(A_i\right) - \prod_{i\in I} P\left(A_i\right) \\ &= \left(1 - P\left(A_1\right)\right) \prod_{i\in I\setminus\{1\}} P\left(A_i\right) \\ &= P\left(A_1^c\right) \prod_{i\in I\setminus\{1\}} P\left(A_i\right) \end{aligned}$$

2.

$$\sigma\left(1_{A_i}\right) = \left\{\varnothing, A_i^c, A_i, \Omega\right\}$$

Take $B_i \in \sigma\left(1_{A_i}\right), i = 1, \cdots, n$. $B_i = \varnothing$ for some i , then both $P\left(\bigcap_{i=1}^n B_i\right)$ and $\prod_{i=1}^n P\left(B_i\right)$ are zero. The ramainings follows from 1.

□

Example 1.1.5

Let X_1, X_2, X_3 be independent random variables with

$$P(X_i = 0) = P(X_i = 1) = 1/2$$

Let $A_1 = \{X_2 = X_3\}$, $A_2 = \{X_3 = X_1\}$ and $A_3 = \{X_1 = X_2\}$. These events are pairwise independent since if $i \neq j$, then

$$P(A_i \cap A_j) = P(X_1 = X_2 = X_3) = 1/4 = P(A_i)P(A_j)$$

but they are not independent since

$$P(A_1 \cap A_2 \cap A_3) = 1/4 \neq 1/8 = P(A_1)P(A_2)P(A_3)$$

1.1.1 Sufficient Conditions

Definition 1.1.6 ▸ Independence between conllections

Collections of sets $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n \subset \mathcal{F}$ are said to be ***independent*** if whenever $A_i \in \mathcal{A}_i$ and $I \subset \{1, \ldots, n\}$ we have $P(\bigcap_{i\in I} A_i) = \prod_{i\in I} P(A_i)$.

Lemma 1.1.7

Without loss of generality we can suppose each A_i contains Ω. In this case the condition is equivalent to

$$P\left(\bigcap_{i=1}^{n} A_i\right) = \prod_{i=1}^{n} P(A_i) \text{ whenever } A_i \in \mathcal{A}_i$$

since we can set $A_i = \Omega$ for $i \notin I$.

Definition 1.1.8 ▸ π,λ- systems

- We say that $\mathcal{A}$ is a π-***system*** if it is closed under intersection, that is, if $A, B \in \mathcal{A}$ then $A \cap B \in \mathcal{A}$.
- We say that $\mathcal{L}$ is a λ-***system*** if
  (i) $\Omega \in \mathcal{L}$.
  (ii) If $A, B \in \mathcal{L}$ and $A \subset B$, then $B - A \in \mathcal{L}$.
  (iii) If $A_n \in \mathcal{L}$ and $A_n \uparrow A$, then $A \in \mathcal{L}$.

Theorem 1.1.9 ▸ $\pi - \lambda$ *Theorem*

If $\mathcal{P}$ is a π-system and $\mathcal{L}$ is a λ-system that contains $\mathcal{P}$, then $\sigma(\mathcal{P}) \subset \mathcal{L}$.

Theorem 1.1.10

Suppose $\mathcal{A}_1, \mathcal{A}_2, \ldots, \mathcal{A}_n$ are independent and each $\mathcal{A}_i$ is a π-system. Then $\sigma(\mathcal{A}_1), \sigma(\mathcal{A}_2), \ldots, \sigma(\mathcal{A}_n)$ are independent.

Theorem 1.1.11

In order for $X_1, \ldots, X_n$ to be independent, it is sufficient that for all

$x_1, \ldots, x_n \in (-\infty, \infty]$

$$P(X_1 \le x_1, \ldots, X_n \le x_n) = \prod_{i=1}^{n} P(X_i \le x_i)$$

Theorem 1.1.12

Suppose $\mathcal{F}_{i,j}$, $1 \le i \le n$, $1 \le j \le m(i)$ are independent π-systems and let $\mathcal{G}_i = \sigma(\bigcup_j \mathcal{F}_{i,j})$. Then $\mathcal{G}_1, \ldots, \mathcal{G}_n$ are independent.

Proof. Let $\mathcal{A}_i$ be the collection of sets of the form $\bigcap_j A_{i,j}$ where $A_{i,j} \in \mathcal{F}_{i,j}$. $\mathcal{A}_i$ is a π-system that contains Ω and contains $\bigcup_j \mathcal{F}_{i,j}$, so Theorem 1.1.10 implies $\sigma(\mathcal{A}_i) = \mathcal{G}_i$ are independent. □

Theorem 1.1.13

If for $1 \le i \le n$, $1 \le j \le m(i)$, $X_{i,j}$ are independent and $f_i : \mathbf{R}^{m(i)} \to \mathbf{R}$ are measurable, then $f_i(X_{i,1}, \ldots, X_{i,m(i)})$ are independent.

Proof. $\sigma\left(X_{i,j}\right), 1 \le i \le n, 1 \le j \le m\left(i\right)$ are independent π- systems. Then $\mathcal{G}_i = \sigma\left(\bigcup_j \left(\sigma\left(X_{i,j}\right)\right)\right)$ are independent. Take any Borel set $B \subseteq \mathbb{R}$, then

$$\begin{aligned}\left(f_i\left(X_{i,1}, \cdots, X_{i,m(i)}\right)\right)^{-1}(B) &= \left(X_{i,1}, \cdots, X_{i,m(i)}\right)^{-1}\left(f_i^{-1}(B)\right) \\ &\subseteq \sigma\left(\bigcup_j \left(\sigma\left(X_{i,j}\right)\right)\right)\end{aligned}$$

Thus $f_i\left(X_{i,1}, \cdots, X_{i,m(i)}\right)$ are independent. □

1.1.2 Independence, Distribution, and Expectation

Theorem 1.1.14

Suppose $X_1, \ldots, X_n$ are independent random variables and X_i has distribution μ_i. Then $(X_1, \ldots, X_n)$ has distribution $\mu_1 \times \cdots \times \mu_n$.

Theorem 1.1.15

Suppose X and Y are independent and have distributions μ and ν. If $h : \mathbf{R}^2 \to \mathbf{R}$ is a measurable function with $h \geq 0$ or $E|h(X,Y)| < \infty$, then

$$Eh(X,Y) = \iint h(x,y)\mu(dx)\nu(dy)$$

In particular, if $h(x,y) = f(x)g(y)$ where $f, g : \mathbf{R} \to \mathbf{R}$ are measurable functions with $f, g \geq 0$ or $E|f(X)|$ and $E|g(Y)| < \infty$, then

$$Ef(X)g(Y) = Ef(X) \cdot Eg(Y)$$

Theorem 1.1.16

If $X_1, \dots, X_n$ are independent and have
(a) $X_i \geq 0$ for all i, or
(b) $E|X_i| < \infty$ for all i, then

$$E\left(\prod_{i=1}^{n} X_i\right) = \prod_{i=1}^{n} EX_i$$

that is, the expectation on the left exists and has the value given on the right.

1.1.3 Sums of Independent Random Variables

Theorem 1.1.17

If X and Y are independent, $F(x) = P(X \leq x)$, and $G(y) = P(Y \leq y)$, then

$$P(X + Y \leq z) = \int F(z-y)dG(y)$$

The integral on the right-hand side is called the ***convolution*** of F and G and is denoted $F * G(z)$. The meaning of $dG(y)$ will be explained in the proof.

Proof.

$$
\begin{aligned}
P(X+Y \leq z) &= \int 1_{(x+y\leq z)}\, \mathrm{d}(\mu \times \nu) \\
&= \int \int 1_{(x+y\leq z)}\, \mathrm{d}\mu\, \mathrm{d}\nu \\
&= \int F(z-y)\, \mathrm{d}\nu \\
&= \int F(z-y)\, \mathrm{d}G(y)
\end{aligned}
$$

where $\mathrm{d}G(y)$ is just a notation for $\mathrm{d}\nu$. We apply this notation motivated from the Lebesgue-Stieljes integration. □

Theorem 1.1.18

Suppose that X with density f and Y with distribution function G are independent. Then $X+Y$ has density

$$h(x) = \int f(x-y)dG(y)$$

When Y has density g, the last formula can be written as

$$h(x) = \int f(x-y)g(y)dy$$

Proof.

$$F(x) = \int_{-\infty}^{x} f(x)\, \mathrm{d}x$$

Since everything is nonnegative, from Fubini's theorem, we get

$$\begin{aligned}
P(X+Y\leq z) &= \int\int_{-\infty}^{z-y} f(x)\,\mathrm{d}x\,\mathrm{d}G(y)\\
&= \int\int_{-\infty}^{z} f(x-y)\,\mathrm{d}x\,\mathrm{d}G(y)\\
&= \int_{-\infty}^{z}\int f(x-y)\,\mathrm{d}G(y)\,\mathrm{d}x\\
&= \int_{-\infty}^{z} h(x)\,\mathrm{d}x
\end{aligned}$$

which shows that $X+Y$ has density $h(x)$.

When Y has density g, then

$$\int f(x-y)\,\mathrm{d}G(y) = \int f(x-y)\,g(y)\,\mathrm{d}y$$

□

Example 1.1.19 ▸ The gamma density

The ***gamma density*** with parameters α and λ is given by

$$f(x)=\begin{cases}\lambda^{\alpha}x^{\alpha-1}e^{-\lambda x}/\Gamma(\alpha) & \text{for } x\geq 0\\ 0 & \text{for } x<0\end{cases}$$

where $\Gamma(\alpha)=\int_0^{\infty}x^{\alpha-1}e^{-x}dx$.

Theorem 1.1.20

If $X = \text{gamma}(\alpha,\lambda)$ and $Y = \text{gamma}(\beta,\lambda)$ are independent, then $X+Y$ is $\text{gamma}(\alpha+\beta,\lambda)$. Consequently if $X_1,\cdots,X_n$ are independent exponential (λ) r.v's, then $X_1+\cdots+X_n$ has a $\text{gamma}(n,\lambda)$ distribution.

Example 1.1.21 ▸ Normal distribution

In Example 1.6.2, we introduced the normal density with mean μ and variance a,

$$(2\pi a)^{-1/2} \exp(-(x-\mu)^2/2a).$$

Theorem 1.1.22

If $X = \text{normal}(\mu, a)$ and $Y = \text{normal}(\nu, b)$ are independent, then $X + Y = \text{normal}(\mu + \nu, a + b)$.

1.2 Weak Laws of Large Numbers